

Acclimatisation process of biogas production from tofu industrial wastewater using biofilter in anaerobic baffled reactor (ABR)

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ARTICLE INFO

Keywords:
Tofu industry
Wastewater
Biogas
Biofilter

ABSTRACT

A biofilter is a simple technology used in an anaerobic baffled reactor (ABR) to keep biological solids (inoculum) from being easily carried by the inlet substrate and to shorten the hydraulic retention time (HRT). The principle of a biofilter is to form a biofilm on the packed bed of the biofilter in ABR so that it contains immobilised microorganisms. This study aims to know the performance of the biofilter reactor on biogas production during the acclimatisation process. The results show biofilters shorten the HRT and effectively remove pollutants, increasing biogas production and methane quality. The total solid content decreases by around 44 %, from 0.38 % to 0.17 %. The biogas production during acclimatisation was 1806.41 L and COD removal was 95 %. The biogas composition of CH₄ was 58.05 %, CO₂ 38.23 %, and N₂ 3.2 %. This study provides preliminary findings for further studies on the use of tofu wastewater as a biogas feedstock with different concentrate substrates, which is very useful for sustainability activities and improving the industry to become green.

1. Introduction

Tofu is a product made with soybean, a legume with great nutritional value. It is a favourite food in Indonesia and other Asian populations; it also has an increasing demand in Europe and is gaining popularity worldwide due to its health benefits and price. Its production begins with soymilk obtained from fresh soybeans [1,2]. In Indonesia, most of the tofu production is carried out by small industries called home industries [3]. During the processing of tofu, a large amount of wastewater is produced which has a bad effect on the environment [null]. The tofu industry generated two types of waste, namely solid waste (tofu dregs) and wastewater [4]. From 150 kg of soybean processing to tofu, 210 kg of tofu dregs, and wastewater of approximately 2115.51 kg are produced [5]. Tofu dregs or okara is a by-product that harms the environment and economy. It comes from the soybean production process, which is the residue component of the filtering of the water-soluble fraction in the form of ground soybean during the production of soymilk [6].

However, in Indonesia tofu dregs do not affect the environment as they can be used directly as the main material for human food or animal feed or other materials such as bioethanol, a nitrogen source in fermentation, and to produce other soy-based foods [7,8]. The main problem of industrial tofu wastewater is its direct disposal in bodies of

water without any treatment which is harmful to the environment [5,9]. The owners of the tofu processing plants generally do not have enough capacity to manage their wastewater due to the complexity of the wastewater treatment process [10]. On the other hand, tofu wastewater still has a high nutritional content that can be used as the main material for new products such as biogas as it has a COD range of approximately 1400–11000 mg/L, BOD₅ 431 mg/L, TSS 244 mg/L, and a pH of about 3.47 [11–13]. Furthermore, the high content of BOD and COD in tofu wastewater causes unpleasant odours in the air and surrounding environment and various types of pollution in water bodies (ground, surface, and river) [null]. The small and medium-scale tofu industry commonly does not have space to treat wastewater, such as wastewater treatment plants (WWTP), as the industry presence is clustered and in the middle of the settlement [null]; [11,14]. Therefore, the utilisation of tofu wastewater for biogas is a good solution to prevent environmental pollution and is economically friendly to the industry to replace fossil energy [15]. The substitute for biogas energy is not only for industry activities; it can also be supplied to the community that lives in the surrounding area that does not have access to a natural gas supply for its household energy, such as cooking and lightning [16]. Acclimatisation is required as an initial test to obtain optimal biogas production and quality. Acclimatisation is very important as a pretreatment for

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<https://doi.org/10.1016/j.renene.2024.121519>

Received 26 September 2023; Received in revised form 26 September 2024; Accepted 30 September 2024

Available online 1 October 2024

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microorganism activities in the anaerobic digestion process. Acclimatisation refers to the process of adapting an anaerobic microorganism to a specific environment or operating conditions, such as temperature, pH, or substrate composition, to optimise its performance and stability in the breakdown of organic matter [17–19]. This process is crucial for efficient anaerobic digestion, enabling microorganisms to thrive and produce biogas effectively.

Acclimatisation can be achieved through gradual changes in operating conditions, continuous feeding, and adaptation of the inoculum. This process contributes to stable anaerobic digestion, even under extreme conditions, such as high levels of ammonia [20,21]. Moreover, acclimatisation is essential for the efficient breakdown of organic matter, biogas production, and sustainable energy sources. By adapting microorganisms to specific conditions, acclimatisation improves the overall efficiency and stability of the anaerobic digester [null].

The biofilter is a technology that can be useful for tofu wastewater during the acclimatisation in the anaerobic digestion process [22]. Biofilter technology for tofu wastewater treatment has some advantages, such as effectively removing organic matter, reducing odour, and producing high-quality effluent [23]. The use of a biofilter in an anaerobic digester is to keep the inoculum so that the inoculum is not easily carried away by wastewater and to shorten the residence time of organic matter (HRT) during the digestion process [24]. The use of biofilters has various advantages helpful for water and wastewater treatment; not only can some of the pollutants be removed, but also greenhouse gases (GHG) such as hydrogen sulphate (H_2S), carbon dioxide (CO_2) and methane (CH_4) can be removed. A study by Nguyen et al. [25] found that a biofilter is effective in removing 30 % of total organic carbon (TOC), 50 % of non-methane volatile organic carbon (NMVOC), and 51 % of NH_3 . The effect of methane by biofilter is very slight in decreasing methane (CH_4) by around 6 %. A study by Li et al. (2016) also found that the biofilter also has a significant effect in reducing the content of COD and total phenol in the effluent of coal gasification wastewater (CGW), respectively, 234 mg/L and 14.2 mg/L, they also found that the methane concentration reached 169 ml CH_4 /L/day.

The present study focuses on the effect of biofilter technology on the anaerobic biofilter reactor (ABR) on the production and composition of biogas, especially the quality of methane during acclimatisation as one of the solutions for wastewater treatment for the transition of tofu industry fuels from fossil fuels to zero carbon emissions.

2. Materials and methods

2.1. Location

This study was conducted on a pilot scale reactor using ABR, and the sample was analysed at the Agro-industrial waste management laboratory, Faculty of Agriculture, University of Lampung, Indonesia.

2.2. Experimental setup

2.2.1. Biofilter setup

The types of biofilter used in the ABR are dense plastic bio balls. The biofilter was placed in an unused fruit box and then weighed to ensure

that the weight of each box was the same. The total number of biofilters used in the ABR chamber was 700 balls and 10 boxes of biofilter with a total weight of 3.56 kg (Fig. 1.).

2.2.2. Anaerobic digestion system

The anaerobic digester used in this study was an anaerobic baffled reactor (ABR) with a total volume of 0.927 m^3 (Fig. 2.). The ABR was inoculated with 50 % inoculum and 50 % substrate/tofu wastewater (a 1:1 ratio). The ABR was recirculated for 7 days to stabilise the chamber pH to 7.16. Then, to avoid microorganism metabolism inhibition, the loading rate concentration was 30 kg COD/m^3 per day and carried out periodically For a duration of 26 days.

2.3. Sample collection

2.3.1. Tofu wastewater and inoculum

The tofu wastewater was collected from micro, small, and medium-scale (MSMEs) of tofu industry in Gunung Sulah District, Bandar Lampung City, Lampung Province, Indonesia. The inoculum was collected from the biogas plant in the tapioca industry in Central Lampung Province, Indonesia (Fig. 3).

2.4. Data collection

2.4.1. Initial data

The initial data consisted of the characteristics of tofu wastewater and inoculum.

a. Inoculum

The data was analysed to obtain data on the C/N ratio, TS, VS, Alkalinity, VFA, and pH.

b. Tofu wastewater characteristic

Tofu wastewater was analysed to obtain CODs, COD, TS, TSS, pH,

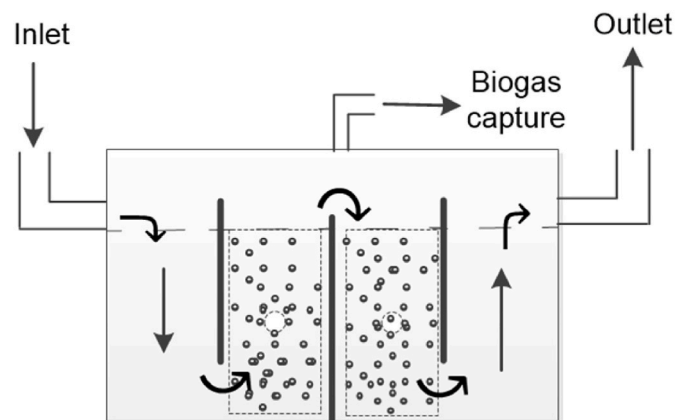


Fig. 2. Design of ABR combined with biofilter.

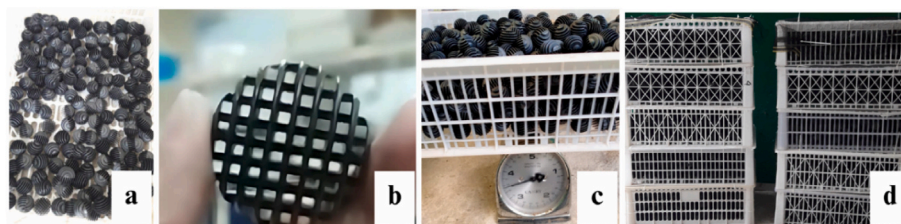


Fig. 1. Biofilter used in the ABR; a) plastic biofilter, b) detail of biofilter, c) biofilter measurements, d) total biofilter used in the ABR.

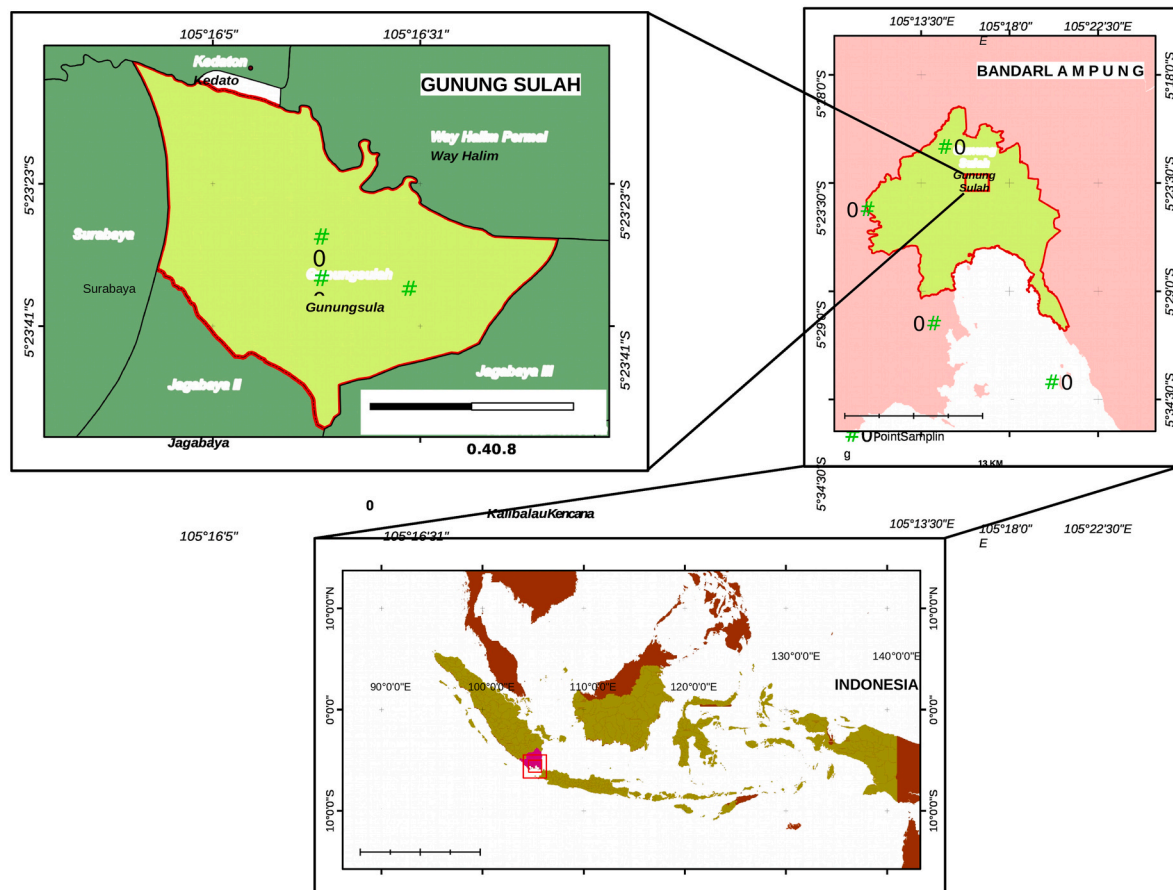


Fig. 3. Tofu wastewater collection location in MSMEs of the tofu industry.

alkalinity, and VFA.

2.4.2. Data analysis

Data for anaerobic digestion were collected to obtain the quality of biogas production, which consists of.

a. Biogas production

Biogas production was observed once every 24 h using a gas flow meter (wet gas flow meter) model W-NK-10 A SINAGAWA.

b. Biogas composition and methane quality

The composition of biogas and the quality of methane were observed to obtain the presentation of the gas composition (CH_4 , CO_2 , and N_2) using gas chromatography (GC) model Simadzu GC-2014 with thermal conductivity detector at temperature 200°C , injection pressure 100 kPa, injection time 1 min, and injection temperature 100°C . The GC was equipped with a shin-carbon column of 4.0 m length and 3 mm inner diameter. Helium was used as a carrier gas with a 40 ml/min flow rate.

3. Results and discussion

3.1. Characteristics of tofu wastewater and inoculum

The characteristic of the nutritional content of tofu wastewater and inoculum is very important due to the effectiveness of the feedstock in the anaerobic digestion process to produce biogas and improve the quality of biogas. Generally, the inoculum is a microbe or strain species selected to improve the transformation of microbes. The inoculum used in this study was taken from the second pond of a biogas outlet in the

tapioca industry, which produces a very good sludge as a seed for the anaerobic reaction, as shown in Table 1.

Inoculums are known as starter cultures to produce fermented products in the agri-food industry, such as cheese, yogurt, etc. In anaerobic digestion, the main sources of inoculum are anaerobic sewage sludge, manure, and digested from agricultural biogas plants [26]. The inoculum is a crucial factor in the initiation of biogas production and quality [27]. The best inoculum for the anaerobic digester method is the outlet of an anaerobic reaction, such as digested sludge [28].

Inoculums from agro-industry such as the tapioca industry wastewater are rich in carbohydrates and sugar, as well as sugar mill wastewater, which has a very good inoculum for anaerobic digestion and also has a high content of organic matter that can potentially be used for fertigation [29,30]. One of the key factors that affect the performance of anaerobic reactors is the content of macronutrients and trace elements in the inoculum and substrate [29].

Furthermore, the result of the analysis of tofu wastewater before anaerobic digestion shows that it has a high COD content and does not comply with the standard of Indonesian government No.15. 2014 for direct discharge into the environment, the results of this analysis are

Table 1
Characteristics of the inoculum.

Parameter	Amount	Unit
pH	7.95	–
C/N ratio	4.66	%
TS	39900	Mg/L
VS	1188.25	mg/L
Alkalinity	800	mg/L
VFA	1224	mg/L

supported by several previous studies which are shown in Table 2.

Not only has high COD content, tofu wastewater also has a low pH which can harm the environment. However, tofu wastewater still has a high organic compound and nutrients such as total sugar, N-total, nitrate, and total Phosphorus which can be reused and is profitable ([33]; S. K. [34]). The utilisation of tofu wastewater in biogas is a good solution that prevents water pollution and replaces fossils as the main energy in the industry for the tofu production process. In addition to that, biogas effluent can be used as organic fertiliser, which has high economic value, so it can be an additional income for the tofu industry [35].

3.2. Production and quality of biogas during acclimatisation

Acclimatisation in anaerobic digestion is a process of adaptation of bacteria that have been transported to tofu wastewater to be processed into biogas. The acclimatisation process is also known as the "seeding process", adding the substrate concentration starts with a small dosage of loading rate up to the actual loading rate to be processed [36]. The concentration of the initial loading rate should start in the low so that the growth of microorganisms will not be overloaded, the reason being to make the up-flow velocity of gas and liquid become low, then the growth of granular and flocculent is encouraged [37]. The working system of the biofilter in ABR is the same as that of the biofilter system in general. The gas is passed through a stack of biofilters where the pollutant decomposer microorganism is immobilised as a biofilm [38].

The accumulation of biogas production during the acclimatisation term is a complex process that involves the adaptation of microorganisms to the substrate and environmental conditions. The anaerobic digester used in this study is ABR combined with a biofilter, also known as an attached growth digester. The HRT in the acclimatisation period is 26 days, and the substrate dosage is 30 L/day with an organic loading rate (OLR) concentration of 0.40 g COD/L.day⁻¹, resulting in the optimal accumulation of biogas production during the acclimatisation period of 1806.41 L. The ability of the digester to adapt efficiently to new substrates, including the types of substrate and the concentration, is affected by the rate of acclimatisation impacts [39]. Biogas production strongly depends on growth supplements and certain compounds that can enhance biogas production [40].

The effect of biofilters on the accumulation of biogas production during the acclimatisation term is a complex process as it can influence the balance between biomass accumulation and removal [41]. The three phases of the biofilter are: the start-up (days 0–25), stable operation (days 26–80), and clogging (days 81–105), with performance of removal and biomass accumulation, respectively, 2.3 kg, 1.1, and 0.5 kg

Table 2
Characteristics of tofu wastewater (control).

Parameter	Unit	Laboratory analysis	Indonesian Government Standard No.5. 2014	Reference		
				1)	2)	3)
BOD	mg/L	–	150	924.97	2369	580
COD	mg/L	12400	300	9900	9350	5759
CODs	mg/L	7150	–	–	–	–
N-Total	mg/L	–	25	673.01	676	–
TSS	mg/L	1188.25	200	–	19888	552
TS	%	0.38	–	–	–	–
pH	–	5.17	6.00–9.00	3.76	6.77	3.9
Alkalinity	mg/L	280	–	–	–	–
VFA	mg/L	1500	–	–	–	–

Sources: 1) [31], 2) [32], 3) [13].

biomass/m³ filter bed/day [42]. Biofilters in biogas can facilitate the removal of gas such as H₂S from biogas, which is essential for its use as an energy source; however, the concentration of H₂S can be increased with increasing biogas production. During acclimatisation on day 6, the concentration of H₂S reaches 45.24 ± 0.01 mL, higher than 20.50 ± 0.70 ml in the non-acclimatisation treatment [43]. Waste-activated sludge during the acclimatisation period can increase the CH₄ production rate by approximately 0.45 m³/day kgCOD removal at an OLR ~20 kgCOD/m³/day on the 6th day of HRT. This indicates that the accumulation of biogas production during acclimatisation can be significant, with a substantial increase in the methane content. As shown in Fig. 4 the increase in the accumulation of biogas production starts on day 8 and continues to day 26, increasing from 466.80 to 1806.41 L/day.

From day 1–7, there is reduced biogas production due to the lag phase; in this phase, the microorganism must adapt to the new condition; this phase is based on the presence of nutrients in the substrate that are ready to use. Additionally, cumulative biogas production is influenced by several factors such as pH, temperature, type of substrate, and C/N ratio; these factors will affect the microorganism in the anaerobic digestion process [44]. Three phases of cumulative biogas production are described, known as the 'sigmoidal curve', namely the lag phase, growth, and asymptotic phases [45].

Biogas is a renewable energy whose main products are methane (CH₄) and carbon dioxide (CO₂), as well as hydrogen sulphur (H₂S) and nitrogen (N₂), which are formed through an anaerobic digestion process [46]. The main gas produced from biogas is CH₄, as it has an economic advantage that can be used as an energy source and to operate a treatment plant [47]. The composition of the biogas consists of 60–70 % methane, 30–40 % CO₂, 1–2% nitrogen, and 1000–3000 ppm H₂S [38]. Our findings showed that the use of biofilters has not only affected the quantity but also the quality of biogas (Table 3). Based on Table 3, the concentration of CH₄ in the acclimatisation term is 58.05 % and the production of CH₄ is 0.147 L/g-CODremoval. However, the production of CH₄ in this study indicated that it was lower than the theoretical production of methane (0.35 L/g-COD removal).

This is because several factors affect the production of biogas and CH₄ during acclimatisation: the dissolution of CH₄ in the wastewater during the anaerobic digestion process and the influx of oxygen due to a leaky reactor. The acritical factor is the types of substrates that are used in anaerobic digestion as different methane potential of carbohydrates and proteins as the main sources of metabolism of acetogenic bacteria will result in different methane production (0.373 vs. 0.417 L/g) [48, 49].

Temperature and OLR concentration also affect methane production; the highest temperature is 33.4 °C and OLR is 3.072 g COD total/L.day results in a methane yield of 0.167 L/g COD total [48]. Although methane production does not reach or approach the standard COD stoichiometry that converts to methane, overall, biogas production and methane concentration results are enhanced during the acclimatisation

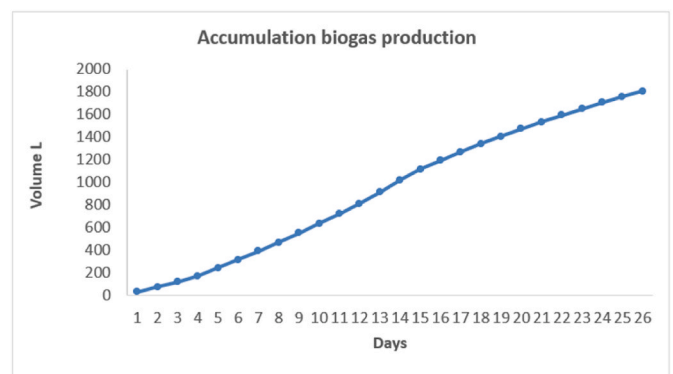


Fig. 4. Accumulation of biogas production during acclimatisation.

Table 3
Biogas composition during the acclimatisation.

Parameter	Amount
CH ₄	58.05 %
CO ₂	38.23 %
N ₂	3.2 %

period in ABR with a biofilter. The conversion of tofu wastewater into biogas can reduce CO₂ by approximately 1.12 kg CO₂e/kg of soybeans and produce CH₄ 0.056 m³/kg of soybeans. It can also substitute 2.82 % of firewood, 11.86 % of LPG, and 33.39 % of biopellets, and can save fuel costs [50].

The result of this study shows CH₄ concentration is high enough for ignition. However, there is no specific standard for CH₄ concentration for the ignition due to the variation depending on the specific application and the condition used, generally, at least CH₄ concentration of 30 % is needed for combustion [51,52]. Moreover, catalytic combustion gas requires a minimum CH₄ concentration of 25 % [53]. Furthermore, the concentration of CH₄ under 5 % gas mixture is too low and not enough for ignition, a gas mixture greater than 15 % is too high and lacks oxygen, however, a concentration between 5 and 15 % is sufficient for combustion [54]. It can be indicated that the concentration of CH₄ in this study is high and can be used for combustion.

The use of biofilter in ABR as a simple technology has merits and demerits in its application. Its merits include the resistance to organic and hydraulic shock, effective degradation of the organic compound and biogas production, a fixed-bed biological reactor with one or more filtration chambers in series, and can be used for wastewater treatment [55–57]. The demerits of biofilters in ABR must also be taken into account, these include, the inability to reduce pathogens and nutrients to an acceptable discharge level, a long start-up process in the absence of adapted seed sludge [18,58,59]. Overall, based on the study results, ABR with a biofilter is very meritorious because it produces high CH₄ concentration and high biogas production, thus it can be considered for the tofu industry owners and related stakeholders in utilising tofu wastewater as a biogas feedstock. However, both the merits and demerits of the ABR technology must be duly considered and reviewed when deciding to implement this method, as they may affect the final outcome.

3.3. The effect of gas and liquid temperature in digester

The general principle of a biofilter is to form a biofilm on the packed bed so that it contains immobilised microorganisms [47]. The use of biofilters in anaerobic digestion is common in reducing some of the composition of biogas, such as H₂S and CO₂; reducing these compounds will improve the quality of biogas [38]. Biofilters affect the processing of biogas production by shortening HRT as the microorganisms grow and attach to the bio ball [28], and the wastewater flows up and downward through the column of the biofilter. Therefore, the contact between the microorganism and the substrate becomes more effective. Afrizal et al. [60] reported that the effect of using a biofilter with the addition of FeCl₃ in the anaerobic digestion of tofu wastewater can reduce the concentration of pollutants, namely, COD removal 94.09 %, TSS removal 94.02 %, and the biogas produced is 8.190 ml.

The transition from using non-renewable energy to renewable energy in industry, transportation, and agriculture will be supported by the European Green Deal Investment Plan, the new flagship initiative of the. Transition activities are under the "Green Deal", countries that are expected to reduce overall emissions by around 50–55 % by 2030 and achieve the 10-year term target [61]. Factors such as HRT, pH, and temperatures influence biogas production. These have an effect not only on the activities of microorganisms but also on the productivity and quality of biogas. The anaerobic microorganism is very sensitive to the pH and temperatures of the substrates. The range temperature of liquid

and gas in this study was 29.15 °C - 29.70 °C. The temperature interval will be different depending on the types of wastewater as the production of the process in each industry is different [62]. The temperature of liquid and gas in ABR is not much different because the weather temperature in Indonesia is relatively constant year-round, both at night and during the day, about 26.8–27 °C (Fig. 5).

The relationship between digester temperature and biogas is very closely related to the anaerobic digestion process, which will affect the quality of biogas; an increase in the temperature in the digestion process will increase the quality of biogas [63]. Temperature is the most important parameter for the biological model that can be affected by the metabolism of microorganisms in the digester; the higher the temperature, the faster the development of biogas [64]. The effect of temperature on the microbial substrate is the absorption kinetics rate at a stable gas formation rate [65].

The low temperature affects the anaerobic digestion process. Low temperatures harm anaerobic digestion. With a decrease in the temperature of the anaerobic process, the result is an increase in the viscosity and solubility of the liquid of gas compounds while reducing the diffusivity of the dissolved substrate [66]. There are advantages and disadvantages of the effective temperature effect in the anaerobic digestion process, such as the activities of microorganisms to form and consume acidity, and the fermentation process is not balanced [67]. The mesophilic temperature is very ideal for the anaerobic digestion process because it has a significant effect on quality, especially methane and productivity biogas, in kinetics and conversion [63]. Temperature significantly affects the anaerobic digestion process, and higher temperature will enhance the dynamic of microbial growth to allow waste to be more optimally converted into biogas [68].

3.4. The efficiency of TS and VS removal

The advantage of using a biofilter for an anaerobic digester is that the inlet substrate does not easily carry biological solids, and the HRT is shorter. The high concentration of biological solids allows for treating the high-strength of soluble wastes [69]. The recommended TS content for food waste treatment is 15–20 % due to its suitability for hydrolysis and methane production [70]. The use of biofilters for anaerobic digestion was reported by Nguyen et al. [25]. Combining biofilter and scrubber significantly reduces greenhouse gas emissions but does not significantly reduce methane concentration in waste gas. The efficiency of the cleaning process through the anaerobic filter is 80–95 %, which can be an option for wastewater treatment plants (WWTP) [71].

The use of biofilters in ABR efficiently removes the solid content of biogas effluents such as TS and VS. The TS/VS ratio is a crucial parameter in the anaerobic digestion process, serving as an indicator of the organic content in the substrate (C. [72]). As shown in Fig. 6 the average of the TS biogas effluent is 0.17 %, and the efficiency of (TS

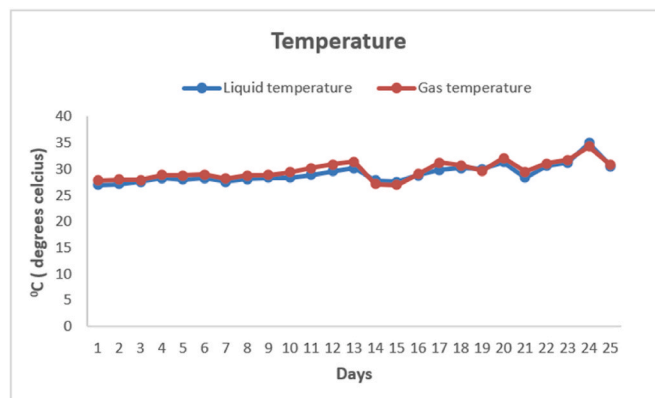


Fig. 5. Temperature of liquid and gas in ABR.

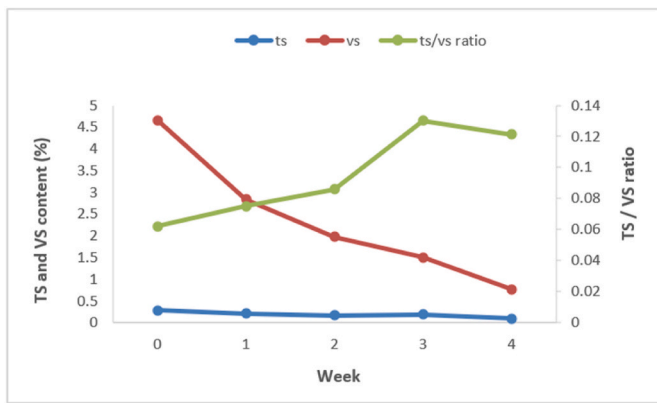


Fig. 6. TS/VS ratio of biogas effluent in acclimatisation term.

removal) is 50 %. Fig. 6 also shows that the TS/VS ratio of biogas effluent gradually increased from 0.06 in the first week up to 0.08 % in the third week, but in the 4th week, the ratio slightly decreased to 0.12. A higher TS/VS ratio indicates a higher organic content, which can be affected by the anaerobic digester process, whether wet or dry.

The TS/VS ratio determines the ability of the substrate to produce biomethane under anaerobic conditions, with a higher ratio related to the COD/VS ratio indicating a higher potential for biogas production [73,74]. The TS/VS ratio also affects the stability of the effluent from the digester; the VS/TS ratio of 0.54 is in HRT on day 7, reducing organic content <45 % [75]. Based on Fig. 6 the highest TS/VS ratio is 13 % in the third week with TS and VS removal, respectively 0.2 and 1.5 %. However, the results of this study show that the ratio of TS/VS is small as the substrate used is tofu wastewater with almost no solid content.

Biogas production depends on the types of feedstocks that are being fermented. Generally, the concentration of solid sewage sludge digestion is approximately 8 and 10 % [71]. Therefore, based on the results of the total solid in Fig. 6 it can be concluded that there is no solid content in the biogas effluent due to the application of the biofilter in the digester. The best combination of technology for tapioca wastewater treatment in COD removal is the Anaerobic-Aerobic Fixed Bio-filter (2F2B) with bee-nest shaped biofilter which contains the indigenous bacterial consortium, and the efficiency of TS removal is 86.2 % [76].

Some factors in this study affected the TS of biogas effluent, namely: I) the types of substrates, and tofu wastewater that contains a small amount of TS that is 0.38 %. II) Filtration before loading substrates into the ABR. The first step before loading the substrate into the ABR is filtering to make retention time shorter and anaerobic digestion faster. III) The solid is stuck in the biofilter and forms a biofilm. Controlling and maintaining biomass on the filter surface is the key point in the use of biofilters.

There are three phases of the pollutant removal mechanism by biofiltration, namely solid, liquid, and gas. Attached biomass that grows around the filter medium is an important factor in the efficiency of the biofiltration system [77]. The operation of biofilters depends on microbial activity; for consistent and effective operation, a constant source of substrate is required [23]. The biofilter system uses the activities of microorganisms to improve the quality of water or air by biological oxidation of various organic substances. There are three factors of biological process in biofilters for wastewater treatment; the attachment of microorganisms in the biofilter, growth, decay, and detachment of microorganisms of microorganisms [78].

3.5. The efficiency of COD removal in acclimatisation term

The use of biofilter in ABR during the acclimatisation terms affects not only biogas production and quality, removal of the TS and TS/VS

ratio of biogas effluent but also COD removal. COD removal is a crucial process in wastewater treatment, and acclimatisation plays a significant role in optimising this process. Acclimatisation involves adapting microorganisms to specific conditions to enhance their ability to remove pollutants, including COD [79]. Based on Fig. 7, COD removal efficiency is 95 %, a decrease from 12400 to 100 mg/L. At first, COD removal is not much, around 61.67 %, as the microorganism still does not show much growth on the surface of the biofilter.

However, over time, the efficiency of COD removal increases by up to 95 %. The acclimatisation process also refers to the adaptation of microorganisms and growth in the media filter to attach and form the biofilm; usually, stable conditions can be achieved after an acclimatisation process of approximately 30 days from the perspective of COD removal [80]. The efficiency of COD removal during acclimatisation term varies depending on the specific conditions and system used. The efficiency of COD removal in the bio-electrochemical system (BES-UASB) is 82 %, with biogas production yields of approximately 0.33 m³/kg-CODremoved, and biomethane production greater than 60 % [81]. To achieve high efficiency of COD removal during the acclimatisation period, several operating conditions have been identified as optimal, such as influent COD content, biomass concentration, and (food/microorganism) F/M ratio, and hydraulic retention time (HRT) [82].

Acclimatisation can also affect the initial phase of COD removal; for instance, wastewater treatment with the optimal glycogen accumulation organisms (GAO) at 21.34 % results in a COD removal rate of 90.2 % [83]. Thus, the evaluation of the capacity to remove COD is essential. According to Ya'acob et al. [84], the treatment of municipal wastewater using an acclimatised mixed culture reaches the optimum of COD and N removal, respectively, 70.41 % and 64.29 % achieved at 150 rpm with HRT for 5 days. Another important key in COD removal of this study is the use of biofilter, which means biofilters are effective in removing COD from wastewater. According to various studies, the efficiency of COD removal by biofilter is influenced by factors such as HRT, temperature, and OLR. The use of biofilters filled with fly ashes from sewage sludge thermal treatment (FASST LWA) as an effective method for airport de-icing wastewater, obtained the highest N removal reached at the temperature of 0 °C (34.93 ± 4.54 %) and hydraulic at 5.0 L m⁻² d⁻¹, however, the most efficient in COD removal is 82.35 % ± 1.53 % at temperature 25 and hydraulic loading of biofilter to 10.0 L m⁻² d⁻¹ [85].

In general, acclimatisation is a critical factor in improving COD removal in wastewater treatment, and understanding its effectiveness in COD is essential to optimise treatment processes.

4. Conclusion

The use of biofilters in ABR for tofu wastewater treatment

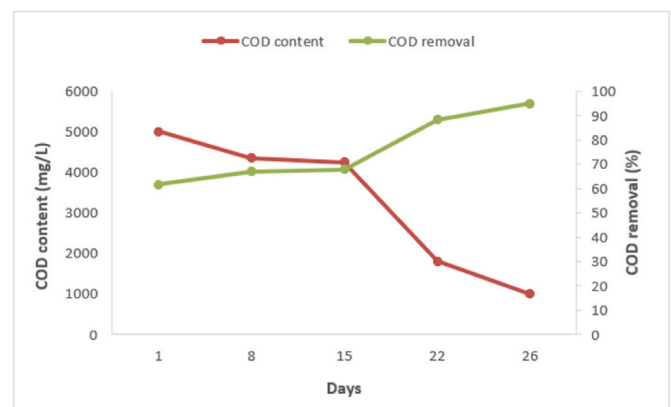


Fig. 7. COD removal during the acclimatisation term.

significantly increases the production of biogas and methane quality, respectively, 69.48 L/day and 58.05 %, and the efficiency of COD removal is 95 %. The operation of the biofilter is based on the micro-organism that sticks to the surface formed by the biofilm to convert it into bioenergy that very effectively removes the total solid content of effluent biogas of approximately 44 % (from 0.38 % to 0.17 %) with TS/VS ratio 0.13 which can be useful for utilising the effluent as a liquid organic fertiliser material. The advantage of using a biofilter in anaerobic digestion is that it makes the HRT shorter and more economical due to the density of the inoculum and the substrate attached to it, so it can be used in the long term. This is highly recommended for industrial-scale wastewater treatment that produces large amounts of wastewater in the production process.

5. Recommendation

The use of a biofilter in ABR is a simple and economically friendly technology that can be used for the long term to prevent the inoculum from being easily carried during substrate loading. Furthermore, biofilter technology is highly efficient in removing pollutants through an anaerobic process and increasing biogas quality. Continuous AD method wastewater treatment with high discharge and shorter HRT is highly recommended.

Funding

This research was funded by the Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, grant number IGA (20243111).

Ethics approval and consent to participate

Not applicable.

Consent to publication

All authors have read and agreed to publish this version of the manuscript.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

CRedit authorship contribution statement

Lydia Mawar Ningsih: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis. **Udin Hasanudin:** Writing – review & editing, Validation, Resources, Methodology, Investigation. **Hynek Roubík:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank the Agro-industrial Waste Management Laboratory, Faculty of Agriculture, University of Lampung, Indonesia, for the support in the data collection process, for establishing an anaerobic digester reactor, and for the possibility of using some of the instruments for analysis. We also thank anonymous SMEs of the tofu industry in Gunung Sulah District, Bandar Lampung City, Indonesia.

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